Data-Driven Modeling of an Undamped Oscillator Using Artificial Neural Networks and Residual Neural Networks

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Machine Learning for Dynamical
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***Abstract*—In this study, the behavior of a nonlinear dynamical system, namely the *Undamped Oscillator 2*, is modeled using data-driven approaches. Two different neural network architectures, a standard Artificial Neural Network (ANN) and a Residual Neural Network (ResNet), are implemented and compared. The dataset is generated by numerically solving the system using a Runge-Kutta method under multiple initial conditions.**

The results show that the ANN model successfully captures the short-term dynamics of the system, while exhibiting error accumulation in long-term predictions. In contrast, the ResNet model fails to provide stable and accurate predictions, diverging significantly from the true system behavior over time. This indicates that the assumptions behind residual learning are not well aligned with the characteristics of the given dynamical system. The study highlights the importance of model selection and hyperparameter tuning in learning nonlinear dynamical systems.

I. Introduction

Modeling nonlinear dynamical systems using machine learning techniques has gained significant attention in recent years. Neural networks provide a flexible framework to approximate complex system behaviors directly from data.

In this study, two architectures are considered:

- A standard feedforward Artificial Neural Network (ANN)
- A Residual Neural Network (ResNet)

The objective is to evaluate whether these models can learn the underlying dynamics of the given system and to compare their performance.

II. Methodology

A. Data Generation

A total of **100 different initial conditions** are generated for the system.

- **80%** of the data is used for training
- **20%** is used for testing

B. Numerical Solution

The system is solved using a **Runge-Kutta-based numerical method**. A suitable time step is selected to ensure both accuracy and stability of the generated trajectories. These solutions serve as the ground truth data.

C. Model Structures

Artificial Neural Network (ANN)
The ANN model is trained to directly map the current system state to the next state.

Residual Neural Network (ResNet)
The ResNet model is trained to learn the residual (difference) between consecutive states, instead of the full mapping.

III. Results and Discussion

A. Loss Analysis

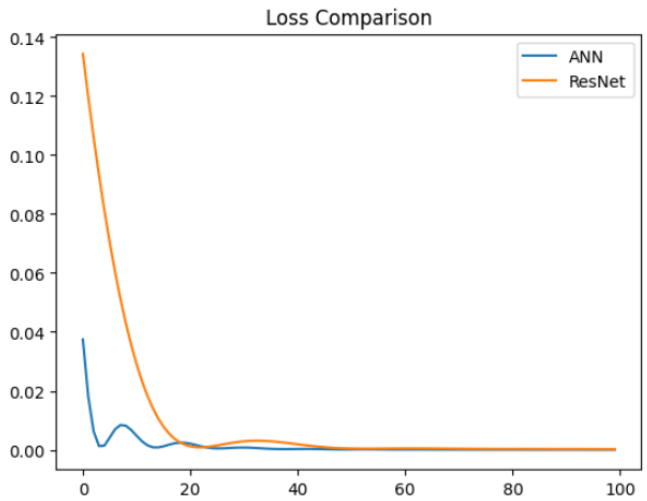

Figure 1. Loss Analysis

As shown in Figure 1:

- The ANN model converges rapidly and stabilizes at a low loss value

- The ResNet model starts with a higher loss and exhibits slower convergence

This confirms that the ANN model achieves better training performance.

B. Trajectory Analysis

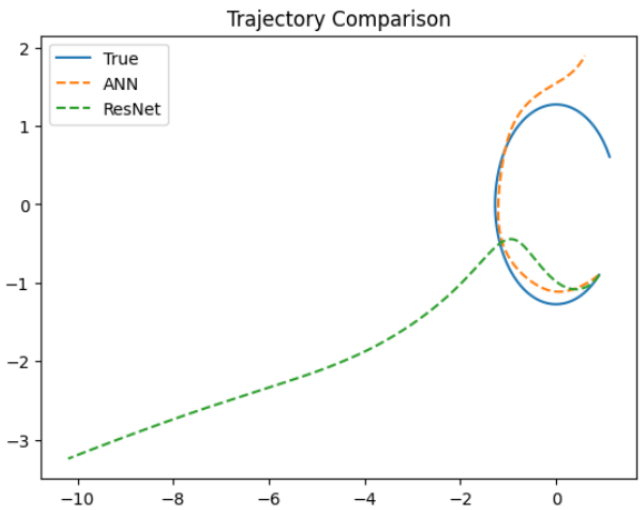

Figure 2. Trajectory Analysis

As shown in Figure 2, the ANN model closely follows the true trajectory of the system. This indicates that the model successfully captures the overall structure of the system dynamics.

In contrast, the ResNet model significantly deviates from the true trajectory. The deviation becomes more pronounced over time, suggesting instability in learning.

C. Time Series Analysis

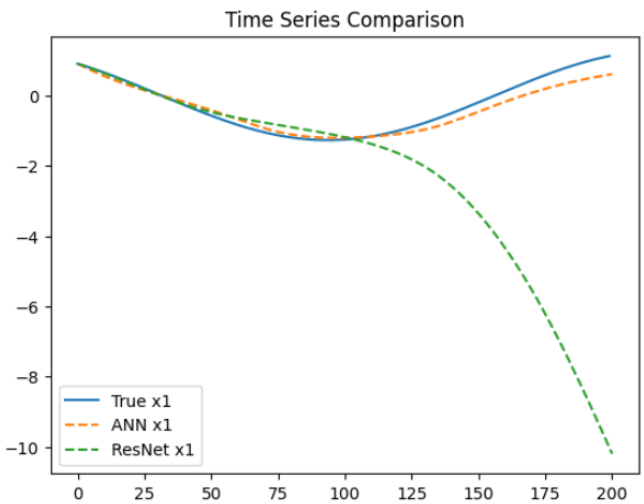

Figure 3. Time Series Analysis

Figure 3 illustrates the time evolution of the system states. The ANN model provides accurate predictions at early time steps, but small errors accumulate gradually.

On the other hand, the ResNet model shows rapid divergence after a certain time point. This behavior indicates that small residual prediction errors compound over time, leading to large deviations.

D. Discussion

The results reveal that the ANN model is capable of learning the short-term dynamics of the system effectively. However, due to its iterative prediction nature, errors accumulate over time, reducing long-term accuracy.

The poor performance of the ResNet model can be attributed to several factors:

- **Mismatch with residual learning assumption:** The system dynamics cannot be effectively represented as small incremental changes.
- **Error accumulation:** Small residual errors grow significantly in iterative predictions.
- **Hyperparameter sensitivity:** Inappropriate learning rate or network depth may lead to unstable training.
- **Lack of physical constraints:** The model does not enforce system properties such as energy conservation, which is critical for undamped systems.

To improve the ResNet performance, techniques such as hyperparameter tuning, data normalization, multi-step training, and physics-informed learning can be considered.

IV. CONCLUSION

In this study, ANN and ResNet models are applied to learn the dynamics of a nonlinear system.

The findings indicate that:

- The ANN model successfully learns the system behavior with reasonable accuracy
- The ResNet model fails to produce stable and reliable predictions

Therefore, it can be concluded that the ANN model is more suitable for this specific problem. Future work may focus on improving model architectures and incorporating physical knowledge into the learning process.